Milestone A: Database I/O Diagnostics - Delivery Report

Date: 12 January 2026

Preview URL: https://science-canvas-creator.vercel.app

Work Completed

Database Logging Infrastructure

1. **Database Logging Table (`dev_io_log`)**: Created table to track all database operations
2. **Supabase Operation Wrapper**: Implemented logging wrapper that intercepts all database calls and logs them to the database (development-only)
3. **Dev Diagnostics Page**: Created development-only page accessible at `/dev/io-diagnostics` with a table listing the last 1000 logs with CSV export functionality

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Problem Identified - Icon Queries

The application was transferring excessive amounts of data when browsing icons, causing slow loading times and timeouts:

Problem 1: Thumbnails Were Full-Size Files

Icon thumbnails stored full SVG files (~200-300 KB each) instead of small compressed images (5-20 KB). This meant browsing icons transferred **50+ MB of data** instead of less than 1 MB.

Problem 2: Database Queries Loaded Too Much Data

- Admin panel fetched all 6,507 icons at once, then displayed only 20
- Canvas editor fetched entire categories (100-300 icons), then displayed only 20
- All queries loaded full icon files instead of just small preview images

Problem 3: Some Upload Methods Didn't Create Compressed Thumbnails

Certain icon upload pathways saved full SVG files as thumbnails instead of generating compressed PNG versions.

Solutions Implemented

1. Created Compressed PNG Thumbnail System

Built a thumbnail generator that converts SVG icons to small 128×128px PNG images with transparent backgrounds, reducing file size by 95%.

2. Updated All Upload Pathways

Fixed all icon upload methods (single file, batch ZIP, AI generation, admin approval) to generate compressed PNG thumbnails automatically.

3. Built Migration Script

Created a script to regenerate thumbnails for all 6,507 existing icons in the database, converting them from full SVG to compressed PNG format.

4. Implemented Database-Level Pagination

Changed queries to load only 20 icons at a time from the database (instead of loading hundreds and hiding most):

- **Admin panel**: Loads 20 icons per page as needed
- **Canvas editor**: Loads 20 icons per category page as needed
- **All queries**: Only fetch thumbnail data for browsing; full icon loads when actually used

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Testing & Verification

Automated Tests (14 tests; 100% passed)

✅ **Icon Library Tests** (6 tests)

- Verified pagination loads exactly 20 icons per page
- Confirmed only thumbnails load initially (not full files)
- Validated full icon loads only when clicked

✅ **Membrane Brush Tests** (5 tests)

- Verified pagination and search limits
- Confirmed full icon loads only when drawing starts

✅ **Upload Pathway Tests** (4 tests)

- Single upload: Creates 4.40 KB PNG thumbnail ✅
- Batch upload: Creates 3.52-3.63 KB PNG thumbnails ✅
- Admin approval: Creates 4.33 KB PNG thumbnail ✅

Manual Testing

- Verified icon quality and appearance
- Tested all user workflows for functionality
- Confirmed transparent backgrounds preserved

Results & Impact

Query Size Improvements (Measured)

Before Optimization:

- Single category (237 icons): 54.8 MB
- Admin page (20 icons): 4.6 MB
- Search (50 icons): 11.5 MB

After Optimization:

- Admin page (20 icons): 66-239 KB
- Search (50 icons): 556 KB
- Category list: 3 KB

Measured Improvement: 95-99% reduction in data transfer per query

Database Optimization

- Total icons processed: 6,507
- Thumbnail size: ~200-300 KB → 5-20 KB per icon

- Total database savings: ~1.2 GB

Summary

- ✅ **Performance Issue Resolved** - Icon browsing now loads <1 MB instead of 50+ MB per page
- ✅ **All Upload Methods Fixed** - Every icon upload pathway creates compressed thumbnails
- ✅ **Existing Icons Migrated** - All 6,507 icons now have optimized PNG thumbnails
- ✅ **Quality Verified** - 14 automated tests passed, manual testing confirmed functionality
- ✅ **Transparent Backgrounds** - All thumbnails preserve icon transparency correctly

The database performance issue has been successfully resolved. Users can now browse thousands of icons without depleting the database Disk I/O budget. All functionality continues to work as expected.
